

## Contact

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## Top Skills

Extract, Transform, Load (ETL)

Microsoft Azure

Python (Programming Language)

## Certifications

React Basics

## Honors-Awards

Winner of Pivot nepal mobile app developers competition - "Birthday Forest"

The Manthan Award - Special mention

## Publications

BLOG: Technology & more...

# Roshan karki

Full Stack Software Engineer ( NodeJS | ReactJS | ReactNative )  
United States

## Summary

Full-Stack Engineer with a 13-year track record of delivering scalable, high-quality software in fast-paced environments. Expert in modern JavaScript frameworks and cloud-native architecture, I bring a unique blend of creative problem-solving and technical rigor to every project. Currently work with technologies including web development, AI engineering, and data lakehouse architectures to build the next generation of intelligent applications.

- 10+ years of full stack software engineer experience in web application development for a diverse clientele, employing the latest technologies in various programming languages
- 4+ years of experience in mobile application development starting from objective C and now to react native for cross-platform mobile application development
- Extensive experience working with database design, implementation in popular RDMS System like PostgreSQL, MySQL along with NoSQL like MongoDB
- Experience managing the entire development life cycle, for various companies worldwide which included requirement analysis, design, development, testing, deployment, documentation and CI/CD
- Extensive experience working with Java Spring Framework, Spring Boot, NodeJS along with Microservice Architecture and Design
- In-depth knowledge of REST and GraphQL api architectures
- Extensive experience working with designing and development of responsive web applications using Vanilla JavaScript, Vanilla CSS, Bootstrap framework, Tailwind CSS framework, HTML5
- Experience with migrating existing applications of various companies to take advantage of the latest technologies
- In-depth knowledge of OOP, functional programming and SOLID Design Principles
- Good knowledge of Git / GitHub and Linux Command Line
- Have good problem-solving skills with excellent written and verbal communication to communicate with different stakeholders of the project with different backgrounds
- Extensive exposure of working in various technical stack with company of different sizes including working for a Canadian based startup company helping them build products from scratch
- Experience of co-founding an IT based startup company which won "PIVOT NEPAL

competition” award from The World Bank in addition to “The Manthan Award South Asia and Asia Pacific” for using information technology and making an impact on society held in New Delhi, IndiaGithub Profile: <https://github.com/rosnk>

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## Experience

### Insulet Corporation

#### Full Stack Engineer

January 2025 - Present (1 year 5 months)

United States

Insulet Corporation (NASDAQ: PODD), headquartered in Massachusetts, is an innovative medical device company dedicated to simplifying life for people with diabetes and other conditions through its Omnipod product platform.

#### Responsibilities:

- >Developed scalable full-stack applications using React, TypeScript, and Modern JavaScript (ES6+) to deliver high-performance user experiences.
- >Contributed to PI Planning and feature refinement sessions to ensure alignment on complex deliverables and high-quality software cycles.
- >Integrated a diverse array of enterprise technologies including Salesforce, Okta, Azure, and Redis to support robust business operations.
- >Engineered custom Python utilities to automate internal processes and meet specific departmental technical requirements.
- >Conducted Root Cause Analysis (RCA) and resolved complex bugs across React frontends, Node.js backends, and Delta Lake synchronization pipelines.
- >Built and maintained comprehensive analytics solutions to provide stakeholders with actionable, data-driven insights.
- >Authored high-quality technical documentation strictly adhering to MedTech industry standards for compliance and auditing.
- >Promoted superior code quality and maintainability through active participation in code reviews and strategic technical debt management.

Technologies: Vite(react), NodeJS, Express, TypeScript, ES6+, Python, Salesforce ORM, OKTA, Azure, Azure devops, Redis, PostgreSQL, ETL, Delta lake, Databricks, REST API, JSON web token, GIT, Postman

### Satisfi Labs

#### Full Stack Engineer (NodeJS, ReactJS )

September 2024 - January 2025 (5 months)

Tampa, FL

Satisfi Labs is a conversational AI platform offering an AI-powered Answering Engine and Live Chat across the web, messaging apps, and voice channels, designed for the sports, entertainment, and tourism industries.

Responsibilities:

- >Designed, developed, and maintained scalable NodeJS (NestJS) web application
- >Build RAG architectural to take advantage of LLM for better contextual answering by utilizing vector databases like pinecone and AlloyDB
- >Used Google Cloud Platform (GCP) services heavily in day-to-day functionality such as AlloyDB, Kubernetes, Artifact Registry, cloud build, cloud function, pub/sub, secret manager and more
- >Implemented Redis as a high-performance Queue Manager for delivering messages from webchat to Core System Application
- >Responsible for UI/UX implementation using reusable components, design systems to sync all the products
- >Collaborated with cross-functional teams (backend, frontend, DevOps) to add new features and bug fixes
- >Participated in daily stand-up meetings as part of following agile Scrum practices for better team coordination and project tracking
- >Conducted thorough testing, including unit and integration tests, to ensure application reliability and performance

ClearDhan LLC

ClearDhan (NodeJS | SpringBoot | ReactJS | ReactNative)

March 2024 - August 2024 (6 months)

New Jersey, USA

ClearDhan is a trading bot Algo that executes rules-based trading strategies based on technical analysis, statistical analysis, AI, ML and mathematical models to make trading decisions.

Responsibilities:

- >Designed, developed, and maintained scalable NodeJS (NestJS) web application for the ClearDhan trading platform, ensuring a smooth user experience
- >Build RAG architectural to take advantage of LLM for better prediction by utilizing vector database like pinecone and alloydb

- >Build a robust streaming architecture for real-time market data ingestion and analysis, enabling low-latency processing of data for ClearDhan's trading platform
- >Utilized artificial intelligence and mathematical modeling techniques to integrate algorithmic trading functionalities within the platform
- >Used google cloud platform (GCP) services heavily in day to day functionality such as alloydb, Kubernetes, Artifact Registry, cloud build, cloud function, pub/sub, secret manager and more
- >Implemented Redis as a high-performance in-memory data store for caching frequently accessed financial data, optimizing the responsiveness and performance of the ClearDhan platform
- >Responsible for UI/UX implementation using reusable components, design systems to sync all the products
- >Collaborated with cross-functional teams (backend, frontend, DevOps) to implement efficient and robust trading solutions
- >Participated in daily stand-up meetings as part of following agile Scrum practices for better team coordination and project tracking
- >Conducted thorough testing, including unit and integration tests, to ensure application reliability and performance

## EKAA Inc

Full Stack Software Engineer (NodeJS, SpringBoot, ReactJS-Microservice Architecture)

October 2020 - December 2023 (3 years 3 months)

Canada, Toronto

"EKaa Inc." is a rental platform that enables renters to rent anything at their disposal and allows borrowers to view and borrow items.

### Responsibilities:

- > Developed and implemented architectural and UI/UX designs, along with database modeling, to ensure robust and scalable solutions
- > Create Software design documentation for the application
- > Designed and implemented a microservice architecture leveraging Eureka for service registry and discovery, including Spring Boot for Java services and NodeJS for Firebase integration
- > Wrote reusable React component along with custom hooks utilizing Typescript for type-safety in frontend (web and expo app)
- > Implemented GraphQL SDL and functionality in backend
- > Implemented various features like authentication, chat, product showcase
- > Integration of third-party API such as Stripe, Algolia, Amazon

- > Collaborated on various Firebase services for both web and mobile applications (EXPO), encompassing phone authentication alongside captcha implementation
- > Participated in daily stand-up meetings as part of following Agile Scrum practices for better team coordination and project tracking
- > Conducted thorough testing, including unit and integration tests, to ensure application reliability and performance

## BLOOMS

### Full Stack Engineer

April 2019 - September 2020 (1 year 6 months)

Singapore

Blooms is a service booking platform based in Singapore with operations in Singapore and Malaysia.

App link:

IOS : <https://apps.apple.com/sg/app/blooms-home-services/id1470036940>

Android : <https://play.google.com/store/apps/details?id=com.blooms.app.v3>

Responsibilities:

Singapore

- > Create and implement Architecture design, UI/UX design, and database design
- > Migrated database from MySQL self-hosted to AWS Aurora (MySQL) for cost optimization, maintainability and scalability
- > Led backend migration from WordPress to Crafter CMS using Spring Boot, improving performance and reliability
- > Architected and developed a RESTful API layer using Spring Boot to bridge the frontend and Crafter CMS, enabling seamless content delivery and management
- > Implemented robust security measures, including role-based access control (RBAC) and content encryption, to safeguard sensitive information
- > Build an android and IOS app and deploy it to the app store
- > Monitored system performance and proactively addressed any issues to maintain website uptime and availability
- > Implemented authentication, CRUD operation, advertisement management features, user roles and more
- > Setup Git and Github for version control, BitBucket for CI/CD

- > Configure AWS EC2 with Ubuntu server, install all the prerequisite for CrafterCMS and setup SSL
- > Communicate with stakeholders on daily standup meeting on the progress of the project following Agile scrum Methodology

### Goingto.do Inc

Full Stack Software Engineer-NodeJS, SpringBoot, ReactJS, React Native

March 2017 - March 2019 (2 years 1 month)

Colorado, USA

"Goingto.do" is a company with a primary focus on digital advertising in the hospitality industry, offering a vast array of solutions to its clients. The nature of work includes analyzing, designing, and developing various web and mobile applications for its clients, including its own event site, goingto.do.

#### Responsibilities:

- > Analysis and transform project requirements into design plans encompassing website design, system design, and database design
- > Come up with UX and UI design and implement it using JavaScript, CSS, Html and CSS responsive framework like bootstrap
- > Utilizing framework like NodeJS (ExpressJS), Spring Framework, React(Create React App) to develop multi-tier MVC based application
- > Develop mobile application in React Native and deploy to both apple and google play store
- > Created reusable component and delivered Single page application using Angular / React in various projects
- > Implemented REST API architecture and endpoint in backend
- > Deploy to EC2/Digital Ocean, debug and maintain sites as per clients requirement

Birthday Forest -Plant trees, Gift trees on occasion like birthday, anniversary OR in memory.

Co-Founder @birthdayforest

March 2013 - February 2017 (4 years)

kathmandu

Birthday Forest is a non-profit company that allows people to plant trees on their birthday or on any occasion. It provides easy access to request a tree plantation through its web and mobile application. This startup was started after winning a prestigious award called "PIVOT NEPAL" organized by the World Bank. Throughout its operation, Birthday Forest has helped thousands

of people to plant trees on their behalf or facilitated them to plant trees on their own.

facebook page link: <https://www.facebook.com/birthdayforest>

#### Responsibilities:

- > Work on web and mobile application development
- > Involved with team members to optimize the application to tailor optimum user experience for planting trees
- > Create vision, strategies, business plan with other co-founders and implement it
- > Create marketing and sales strategies including online marketing, and implement it
- > Tie up with community forest to allocate land for tree plantation
- > Find potential partners in different sectors to increase awareness and visibility about the organization and its offerings

#### Nepaltoday

##### Senior Frontend Developer

January 2012 - February 2013 (1 year 2 months)

Kathmandu, Bāgmatī, Nepal

Nepaltoday is a news portal website having customer base in Nepal and Nepalese diaspora throughout the world.

#### Responsibilities:

- > Conduct user research, design user flows, wireframes and create prototypes design
- > Slice Photoshop design into CSS/html and develop interactive component using bootstrap, SCSS pre processor
- > Convert interactive CSS/html site to WordPress theme
- > Make sure WordPress theme is built with mobile first approach to make it responsive using media query
- > Make sure site follows proper HTML/CSS structure to make site accessible to everyone
- > Communicate closely with backend developers, team lead to communicate Rest API needs.

Technologies: HTML/CSS3, JQuery, JavaScript, PHP, WordPress, Photoshop, Bootstrap, MySQL, XAMPP

## Education

University of Portsmouth

Honours Degree of Bachelor of Science in Computing · (2009 - 2011)